

Pololu 3.3V, 2.6A Step-Down Voltage Regulator D24V22F3

This is a merged information page for Item #2857.

[View normal product page.](#)

Pololu item #: 2857

Brand: [Pololu](#)

Status: Active and Preferred 

 **RoHS3**



Output voltage	Typical max output current ¹	Input voltage range ²
3.3 V	2.6 A	4 V – 36 V

Note 1: At 24 V in. Actual achievable maximum continuous output current is a function of input voltage and is limited by thermal dissipation. See the output current graphs on the product pages for more information.

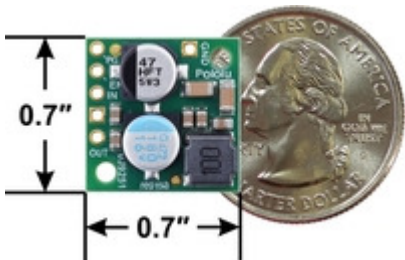
Note 2: Minimum input voltage is subject to dropout voltage considerations; see the dropout voltage section of product pages for more information.

Alternatives available with variations in these parameter(s): output voltage [Select variant...](#)

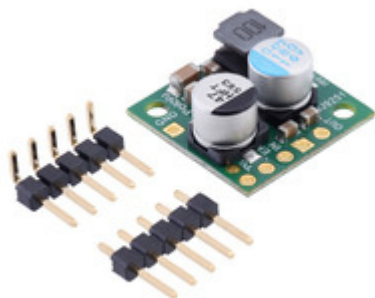
Pictures



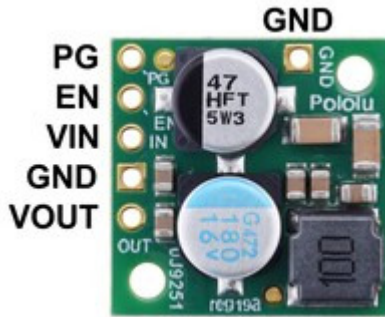
Pololu 3.3V, 2.6A Step-Down Voltage Regulator D24V22F3.



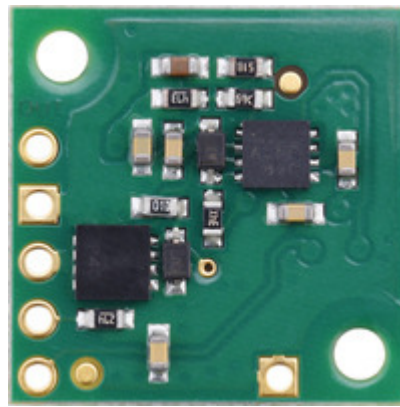
Pololu Step-Down Voltage Regulator D24V22Fx, top view with dimensions.



Pololu Step-Down Voltage Regulator D24V22Fx with included hardware.



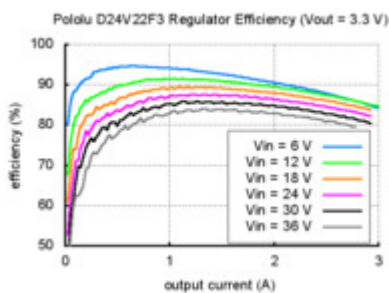
Pololu Step-Down Voltage Regulator D24V22Fx, top view with labeled pinout.



Pololu Step-Down Voltage Regulator D24V22Fx, bottom view.



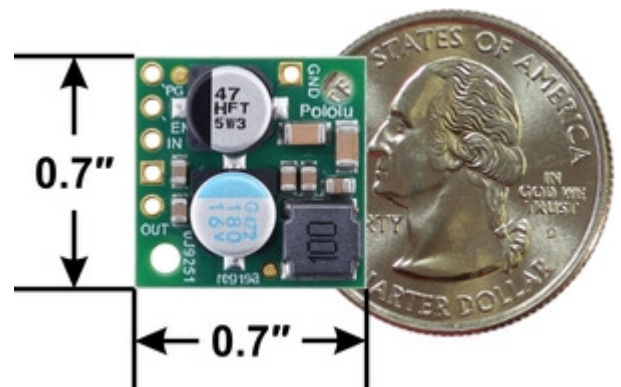
Pololu Step-Down Voltage Regulator D24V22Fx, side view.



Typical efficiency of Pololu 3.3V, 2.6A Step-Down Voltage Regulator D24V22F3.

Overview

The D24V22Fx family of step-down voltage regulators generates lower output voltages from input voltages as high as 36 V. They are synchronous switching regulators (also called switched-mode power supplies (SMPS) or DC-to-DC converters) with typical efficiencies of 85% to 95%, which is much more efficient than linear voltage regulators, especially when the difference between the input and output voltage is large. These regulators can typically support continuous output currents between 1.4 A and 2.6 A, though the actual available output current is a function of the input voltage and efficiency (see the [Typical efficiency](#) and [Maximum continuous output current](#) sections below). In general, the available output current is a little higher for the lower-voltage versions than it is for the higher-voltage versions, and it decreases as the input voltage increases.



These regulators have a typical quiescent (no load) current draw of around 1 mA, and an enable pin can be used to put the boards in a low-power state that reduces the quiescent current to approximately 5 μ A to 10 μ A per volt on VIN.

The modules have built-in reverse-voltage protection, short-circuit protection, a thermal shutdown feature that helps prevent damage from overheating, and a soft-start feature that reduces inrush current.

Several different fixed output voltages are available:

- [**D24V22F3: Fixed 3.3V output**](#)
- [**D24V22F5: Fixed 5V output**](#)
- [**D24V22F6: Fixed 6V output**](#)
- [**D24V22F7: Fixed 7.5V output**](#)
- [**D24V22F9: Fixed 9V output**](#)
- [**D24V22F12: Fixed 12V output**](#)

The different voltage versions of this regulator all look very similar, so you should consider adding your own distinguishing marks or labels if you will be working simultaneously with multiple versions. This product page applies to all versions of the D24V22Fx family.

The D24V22Fx family is intended to replace our older D24V25Fx family of step-down voltage regulators. The two designs have the same size and similar current capabilities and input voltage ranges, but they do not have the same pinout and are based on different internal circuits, so there are fundamental differences in operation. In particular, these newer D24V22Fx regulators have much lower dropout voltages and provide a “power good” signal, and the newer design allows for higher output voltages (e.g. 12 V).

For higher-power alternatives, please consider our [**D36V28Fx family of step down regulators**](#) that can handle current between 2 A and 4 A and take input voltages up to 50 V.

We manufacture these boards in-house at our Las Vegas facility, which gives us the flexibility to make these regulators with customized components to better meet the needs of your project. For example, if you have an application where the input voltage will always be below 20 V and efficiency is very important, we can make these regulators a bit more efficient at high loads by replacing the 40V reverse voltage protection MOSFET with a 20V one. We can also customize the output voltage. If you are interested in customization, please [**contact us**](#).

Features

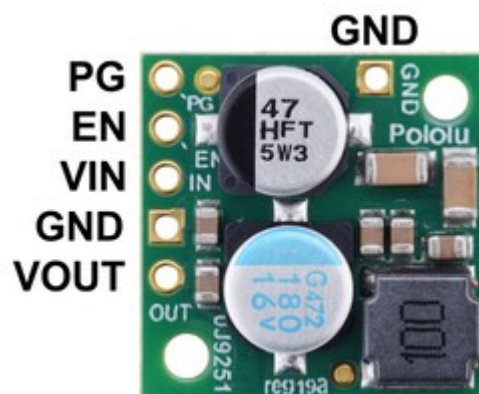
- Input voltage:

- 4 V to 36 V for the version that outputs 3.3 V
- $[output\ voltage + dropout\ voltage]$ to 36 V for output voltages of 5 V and higher (see the [dropout voltage section](#) for details)
- Fixed 3.3 V, 5 V, 6 V, 7.5 V, 9 V, or 12 V output (depending on regulator version) with 4% accuracy
- Typical maximum continuous output currents between 1.4 A and 2.6 A (see the [maximum continuous output current graph](#) below)
- Typical efficiency of 85% to 95%, depending on input voltage, output voltage, and load (see the [efficiency graph](#) below)
- Switching frequency: ~400 kHz
- Integrated reverse-voltage protection, over-current and short-circuit protection, over-temperature shutoff, and soft-start
- 1 mA typical no-load quiescent current; this can be reduced to approximately 5 μ A to 10 μ A per volt on VIN by disabling the board (see the [quiescent current graph](#) below)
- Power-good indicator can be used to tell when the regulator has reached and is maintaining its target output voltage
- Compact size: 0.7" $\times$ 0.7" $\times$ 0.31" (17.8 mm $\times$ 17.8 mm $\times$ 8 mm)
- Two 0.086" mounting holes for #2 or M2 [screws](#)

Using the regulator

Connections

These buck regulators have five main connection points for five different electrical nodes: power good (PG), enable (EN), input voltage (VIN), ground (GND), and output voltage (VOUT). The board also features a second ground connection point off the main row of connections that might be convenient for applications where you are soldering wires directly to the board rather than using it in a breadboard.



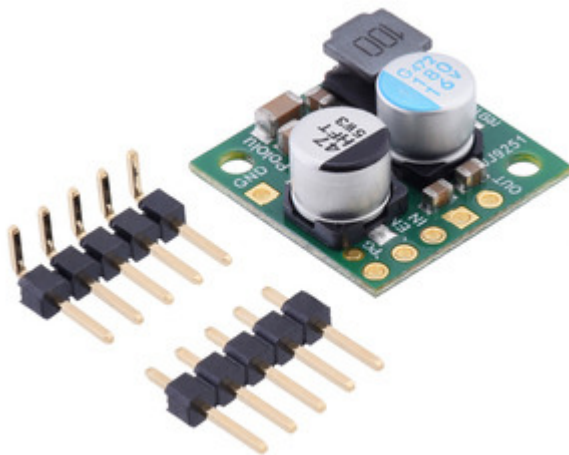
The input voltage, **VIN**, powers the regulator. Voltages between 4 V and 36 V can be applied to VIN, but for versions of the regulator that have an output voltage higher than 4 V, the effective

lower limit of V_{IN} is V_{OUT} plus the regulator's dropout voltage, which varies approximately linearly with the load (see below for a graph of dropout voltages as a function of the load).

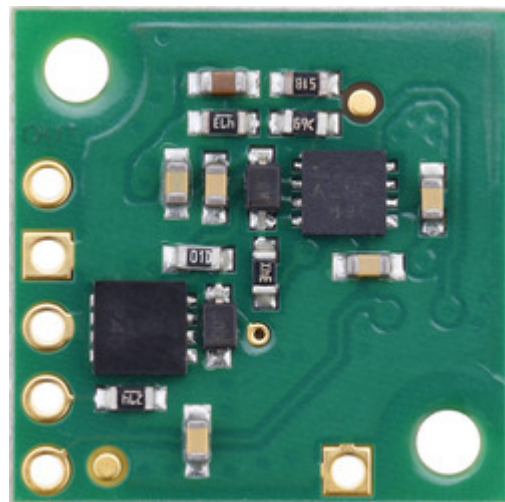
The output voltage, **V_{OUT}** , is fixed and depends on the regulator version: the D24V22F3 version [outputs 3.3 V](#), the D24V22F5 version [outputs 5 V](#), the D24V22F6 version [outputs 6 V](#), the D24V22F7 version [outputs 7.5 V](#), the D24V22F9 version [outputs 9 V](#), and the D24V22F12 version [outputs 12 V](#).

The regulator is enabled by default: a 270 k Ω pull-up resistor on the board connects the **EN** pin to reverse-protected V_{IN} . The EN pin can be driven low (under 1 V) to put the board into a low-power state. The quiescent current draw in this sleep mode is dominated by the current in the pull-up resistor from EN to V_{IN} and by the reverse-voltage protection circuit, which altogether will draw between 5 μ A and 10 μ A per volt on V_{IN} when EN is held low. If you do not need this feature, you should leave the EN pin disconnected.

The “power good” indicator, **PG**, is an open-drain output that goes low when the regulator's output voltage falls below around 85% of the nominal voltage and becomes high-impedance when the output voltage rises above around 90%. An external pull-up resistor is required to use this pin.

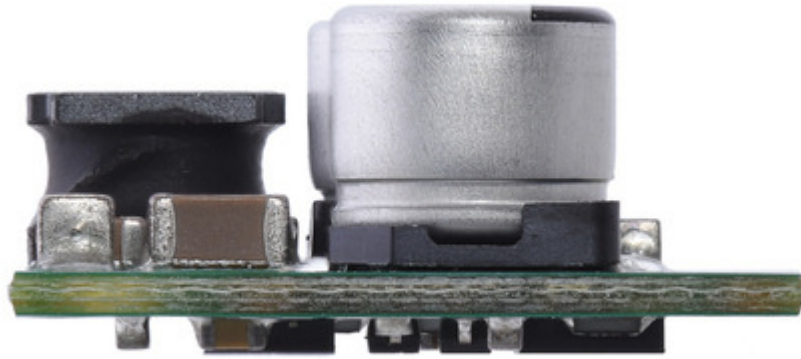


Pololu Step-Down Voltage Regulator D24V22Fx with included hardware.



Pololu Step-Down Voltage Regulator D24V22Fx, bottom view.

The five main connection points are labeled on the top of the PCB and are arranged with a 0.1" spacing for compatibility with solderless [breadboards](#), [connectors](#), and other prototyping arrangements that use a 0.1" grid. Either the included 5×1 [straight male header strip](#) or the 5×1 [right angle male header strip](#) can be soldered into these holes. For the most compact installation, you can solder wires directly to the board.



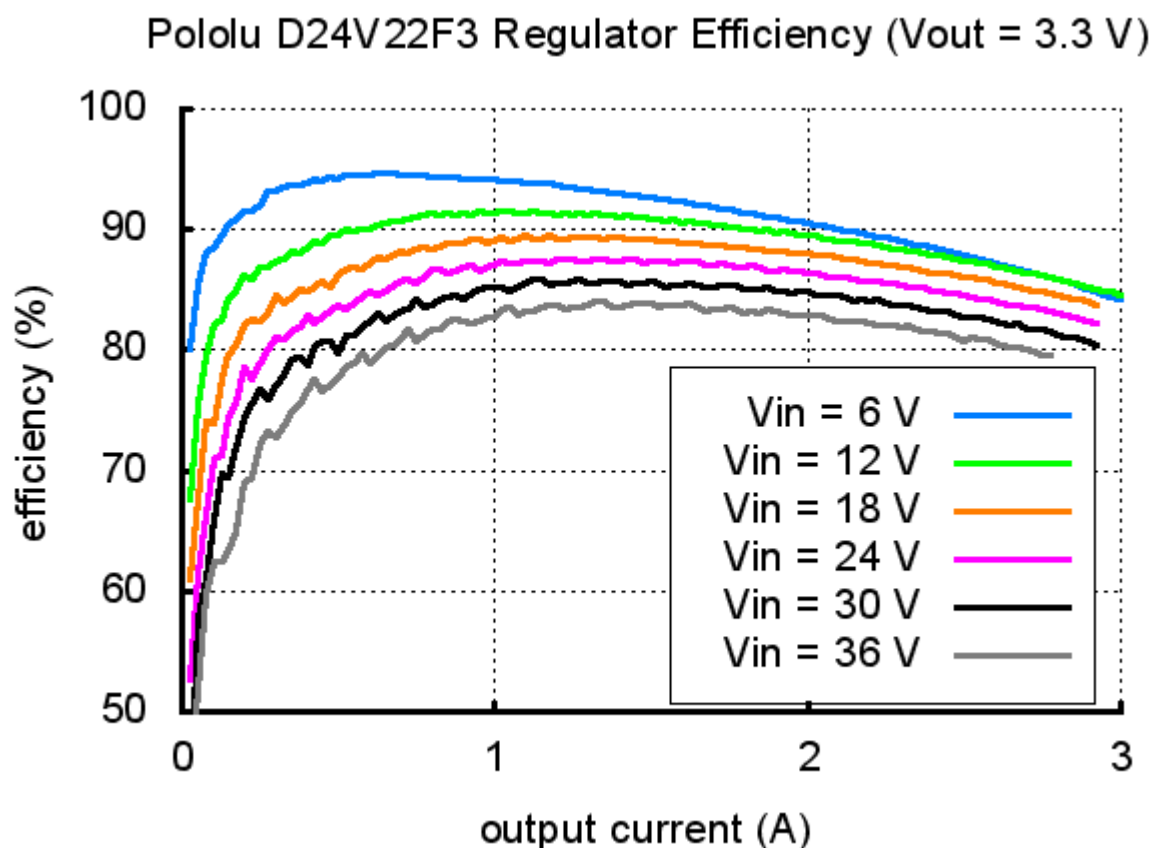
Pololu Step-Down Voltage Regulator D24V22Fx, side view.

The board has two 0.086" (2.18 mm) diameter mounting holes intended for #2 or M2 [screws](#). The mounting holes are at opposite corners of the board and are separated by 0.52" (13.21 mm) both horizontally and vertically. For all the board dimensions, see the [dimension diagram](#) (204k pdf).

Details for item #2857

Typical efficiency

The efficiency of a voltage regulator, defined as (Power out)/(Power in), is an important measure of its performance, especially when battery life or heat are concerns. This family of switching regulators typically has an efficiency of 85% to 95%, though the actual efficiency in a given system depends on input voltage, output voltage, and output current. See the efficiency graph below for more information.

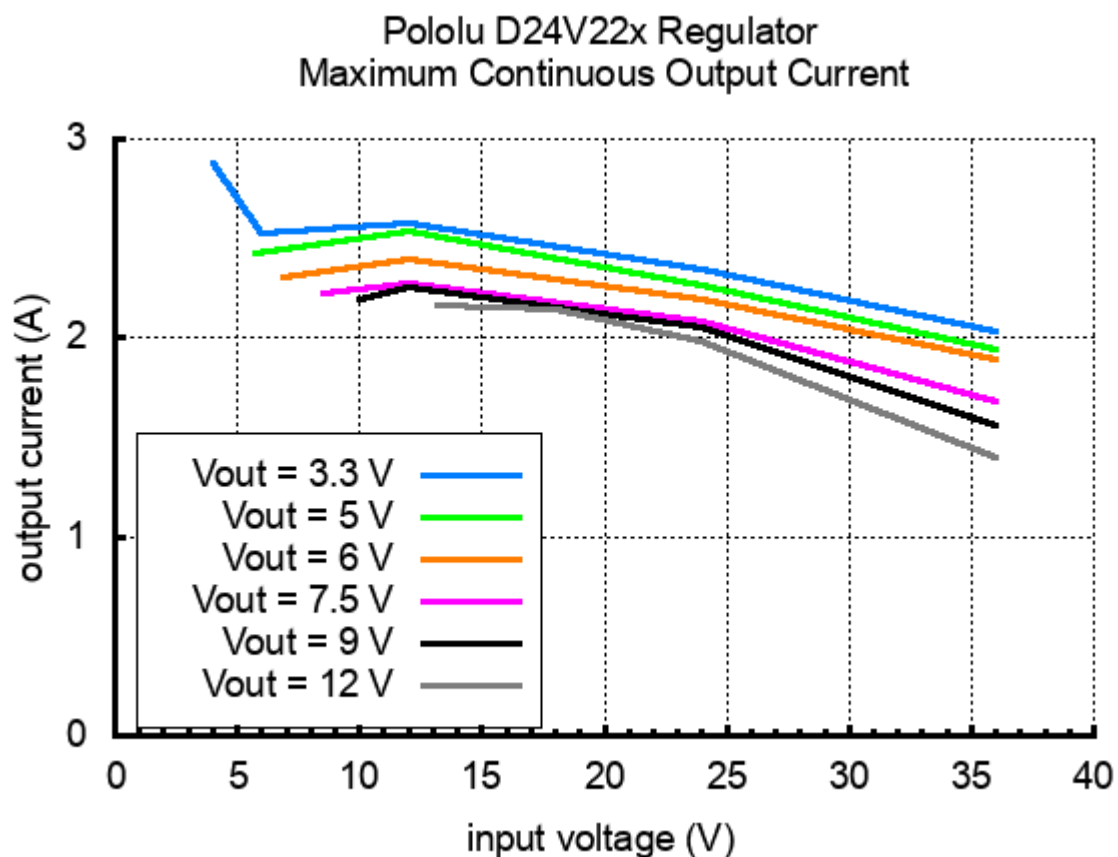


Typical dropout voltage

The dropout voltage of a step-down regulator is the minimum amount by which the input voltage must exceed the regulator's target output voltage in order to ensure the target output can be achieved. For example, if a 5 V regulator has a 1 V dropout voltage, the input must be at least 6 V to ensure the output is the full 5 V. Generally speaking, the dropout voltage increases as the output current increases. Since this family of regulators require an input voltage of at least 4 V, dropout voltage is not a consideration for this 3.3 V version.

Maximum continuous output current

The maximum achievable output current of these regulators varies with the input voltage but also depends on other factors, including the ambient temperature, air flow, and heat sinking. The graph below shows maximum output currents that these regulators can deliver continuously at room temperature in still air and without additional heat sinking.

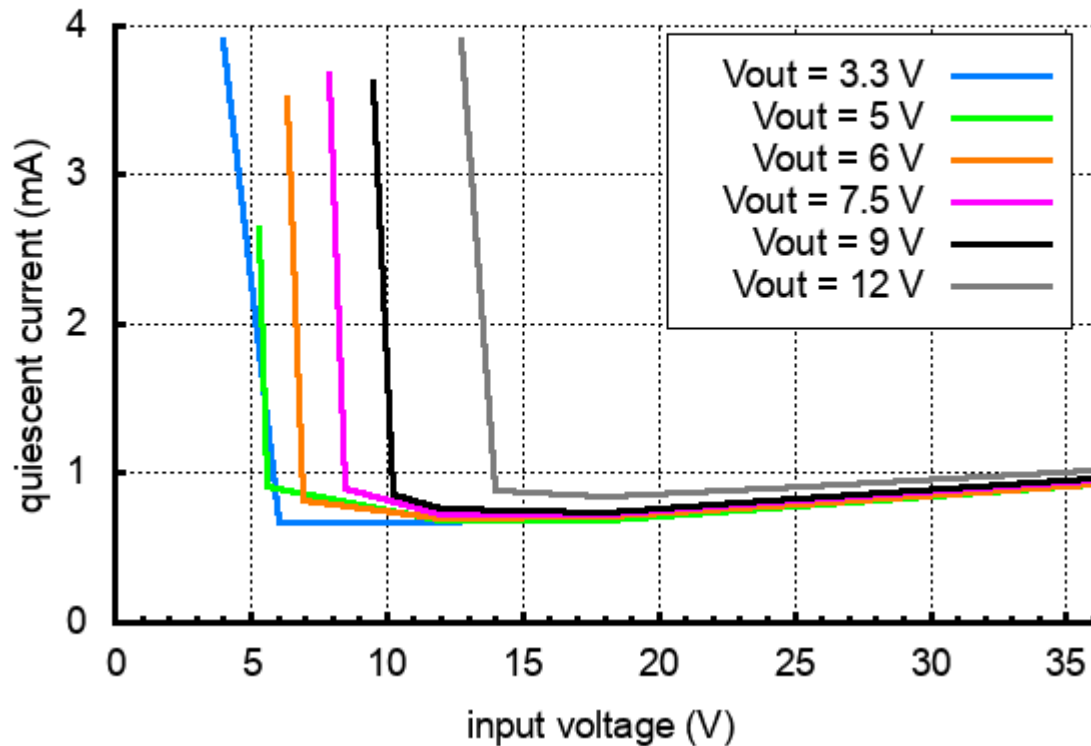


During normal operation, this product can get hot enough to burn you. Take care when handling this product or other components connected to it.

Quiescent current

The quiescent current is the current the regulator uses just to power itself, and the graph below shows this for the different regulator versions as a function of the input voltage. The module's EN input can be driven low to put the board into a low-power state where it typically draws between 5 μ A and 10 μ A per volt on VIN.

Pololu D24V22Fx Regulator Quiescent Current



Dimensions

Size: 0.7" × 0.7" × 0.31"¹

Weight: 2.3 g¹

General specifications

Minimum operating voltage:	4 V
Maximum operating voltage:	36 V
Continuous output current:	2.3 A ²
Output voltage:	3.3 V
Reverse voltage protection?:	Y
Maximum quiescent current:	1 mA ³

Identifying markings

PCB dev codes:	reg19a
Other PCB markings:	0J9251

Notes:

¹ Without included optional headers.

- 2 Typical max continuous output current at 24 V in. Actual achievable continuous output current is a function of input voltage and is limited by thermal dissipation. See the output current graph under the description tab for more information.
- 3 Quiescent current is typically under 1 mA for most of the operating range. The ENABLE pin can be used to reduce the quiescent current to less than 10 μ A per volt on VIN.

File downloads

Dimension diagram of the D24V22Fx Step-Down Voltage Regulator (204k pdf)

3D model of the D24V22Fx Step-Down Voltage Regulator (4MB step)

Drill guide for the Pololu D24V22Fx family of step-down voltage regulators (27k dxf)

Frequently-asked questions

No FAQs available.

On the blog



New products: 3.3V, 5V, 6V, 7.5V, and 9V Step-Down Voltage Regulator D24V22Fx by Ben - 1 December 2015

This post is a few weeks late, but I am following up on my original D24V22F12 12V regulator introduction to let you know that we now

have 3.3 V,...